



Fig. 9: File face1.png, face2.png



Fig. 10: The robot giving introduction

code. Thus, there are many threads in the code, which have been controlled using global flags. The values of the global flags change based on the voice commands given by the user.

The following Python libraries need to be installed in Raspberry Pi for this project using the command:

```
sudo pip3 install library_name
```

OpenCV: Used for image processing based tasks



Fig. 11: The robot solving the puzzle

Numpy: Used for working with arrays

Pytesseract: User for optical character recognition

Pygame: Used to create GUIs

Speech Recognition: Used for speech recognition

Imutils: Used to capture video

For the robot to solve sudoku puzzles, we need some sort of algorithm. Backtracking is used in this situation as it is particularly effective for such applications. The following is a straightforward backtracking strategy for solving a sudoku puzzle:

Find row, col of an unassigned cell

If there is none, return true

For digits from 1 to 9

- a) If there is no conflict for digit at row, col
assign digit to row, col and recursively try fill in rest of grid
- b) If recursion successful,
return true

c) Else, remove digit and try another

If all digits have been tried and nothing worked, return false

First, create the code for face expressions of the talking robot, then the code for talking and speech recognition. Next, create the code for capturing the sudoku picture and processing the image for solving the puzzle. So, the code is created for all of these and saved. The image file for the simple face animation looks as shown in Fig. 9.

Now that the robot is ready, it may be switched on and asked to give its intro before solving the puzzle. To solve sudoku, put the puzzle in front of it and say “capture.” The robot will capture the sudoku puzzle’s image and start solving it, as shown in Fig. 10 and Fig. 11.

Though this robot can only solve sudoku puzzles for now but, due to its powerful processing unit, many other computer-vision based tasks can also be performed using it. So, in future, you may see this robot performing some more interesting and cool things. **EFY**

EFY Note

The source code of this project is available

for free download at www.electronicsforu.com



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